Capacitor Voltage Clamped Series Resonant Power Supply with Improved Cross Regulation

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ABSTRACT

In this paper we present the performance characteristics of a half-bridge Series Resonant Converter with clamped resonant capacitor using the state-space technique. This converter topology has an advantage over the conventional SRC that no energy is returned to the source, therefore, the peak current drawn from the source is limited and the power dissipation in the source impedences in practice is avoided. To varify the analytical results, a two-output power supply was designed and tested. The problem of cross regulation in the two outputs was solved by using a nonlinear inductor as a switch in one of the output circuits. The improvement in performance is illustrated with experimental results.

I. INTRODUCTION

Series Resonant Converter have many advantages such as low converter gain, using capacitive output filter, short-circuit protection and simplicity in control. However, in the converter operation, the energy exchange takes place between the source and the resonant tank. The energy, returned to the source, increases the peak values of the voltage and current in the converter, and causes power dissipation in the source impedence in practice. This problem can be avoided by clamping the voltage across the the resonant capacitor. Such a converter is called the capacitor voltage clamped series resonant converter (SRC-CVC). In this paper we will present the analysis of the converter and the experimental results of a two-output SRC-CVC power supply.

We present the solution for the classic problem of cross regulation which exists in the secondary windings of the output transformer, when only one of the output voltage is regulated by feedback control. This problem arises due to the differential load currents and the unequal leakage inductances in the secondary windings. The proble has been solved by using a current controlled nonlinear inductor in the output circuit as a switch to isolate the two outputs over a part of the switching period . Experimental results show a considerable improvement in the Cross Regulation.

II. ANALYSIS OF HALF-BRIDGE SRC-CVC

The Half-bridge SRC-CVC is given in Fig.1 and its equivalent circuit is shown in Fig.2. In the equivalent circuit, the leakage inductance of the transformer is included in the resonant inductor L. The output circuit, which consists of filter capacitor C_0 , load resistor R_L, and the rectifier circuit can be referred to the primary side of the ideal transformer and is modelled by a Voltage Sink V_E . Since the converter is operated in the discontinuous conduction mode the voltage sink in the circuit may take on three values: $+V_0$, $-V_0$ and open circuit, where V_0 is the output voltage. From Fig.1, we have the following relation,

$$V_E = \begin{cases} \dfrac{i_L}{|\,i_L\,|} V_o & i_L \neq 0 \\ \text{open} & i_L = 0 \end{cases}$$

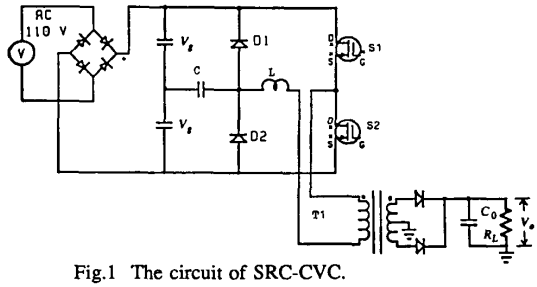

Fig.1 The circuit of SRC-CVC.

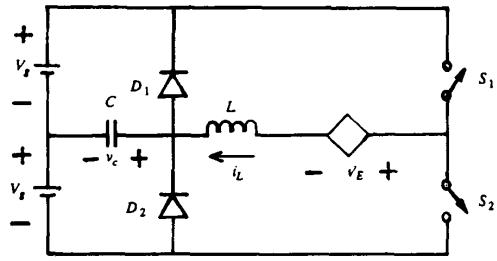

Fig.2 The Equivalent circuit of SRC-CVC.

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In this paper we assume that the switch S_1 and S_2 are driven alternately by a square wave signal of period T_s at 50% duty cycle. The diodes D_1 and D_2 clamp the resonant capacitor voltage to $+V_g$ and $-V_g$ respectively.

2.1 Steady State Response

The steady state operation of the converter in the discontinous mode over a half switching period can be descrived by three circuit modes m_1 to m_3, as shown in Fig. 3(a). The voltage and current responses in each of these circuit modes over the specified time interval is given below. We use normalized quantities in which all voltages and currents are normalized by V_g and V_g/Z_0, respectively, where $Z_0 = \sqrt{L/C}$,
The analysis of each mode is given as follows;

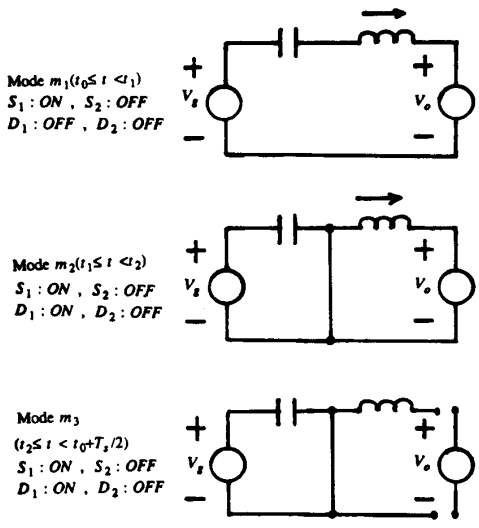

Fig.3(a) Circuit modes in a half switching period.

2.1.1 Mode $m_1(t_0 \le t < t_1)$

The normalized state-plane equation for this circuit mode is given by,

$$\frac{di_{nL}}{dv_{nc}} = \frac{-v_{nc} + (1 - V_{no})}{i_{nL}} \tag{1}$$

where we assume that $i_{nL} = 0$ and $v_{nC} = -1$ at $t = t_0$.

Substituting $V_{no} = M$, it can be shown from (1) that the steady state solution can be represented on the v_{nc}–i_{nL} plane by a directed circular segment from t_0 to t_1 with a singular point $(1-M,0)$. The resonant inductor current i_{nL} and capacitor voltage v_{nc} at any time during the circuit mode are given by,

$$i_{nL}(t) = (2-M)\, Sin\, \omega_0(t-t_0) \tag{2}$$

$$v_{nc}(t) = (1-M) - (2-M)\, Cos\, \omega_0(t-t_0) \tag{3}$$

The capacitor voltage rises until it is clamped at t_1 by the source voltage V_g. The angle subtended by the circular arc in this time interval is given by,

$$\omega_0(t_1-t_0) = Cos^{-1}\left(-\frac{M}{2-M}\right) \tag{4}$$

and from (2) we have,

$$i_{nL}(t_1) = (2-M)\, Sin\, \omega_0(t_1-t_0) \tag{5}$$

also

$$v_{nc}(t_1) = 1.0$$

which serve as the initial condition for the m_2 circuit mode.

2.1.2 Mode $m_2(t_1 \le t < t_2)$

In this circuit mode, the capacitor is clamped at V_g and the inductor L is de-energised by the voltage sink V_0. The normalized voltage and current at any time in the mode are given by,

$$i_{nL}(t) = i_{nL}(t_1) - M\,\omega_0(t-t_1) \tag{6}$$

$$v_{nc}(t) = 1 \tag{7}$$

The steady state trajectory is a vertical line on the v_{nc}–i_{nL} plane. The inductor current drops to zero.

2.1.3 Mode $m_3(t_2 \le t < t_0+T_s/2)$

In this circuit mode, the bridge rectifier presents an open circuit to the inductor and

$$i_{nL} = 0 \quad ; \quad v_{nc} = 1 \tag{8}$$

The inductor current remains zero and the capacitor voltage clamped to V_g until the switch S_1 is opened at $t_0 + T_s/2$. The Mode operation is represented by a point (1,0) on the state-plane.

The steady state response of the converter in the other half switching period can be represented by a trajectory which has the odd symmetry to the origin on the state-plane. The complete state-plane representation of the steady-state converter response is shown in Fig.3(b).

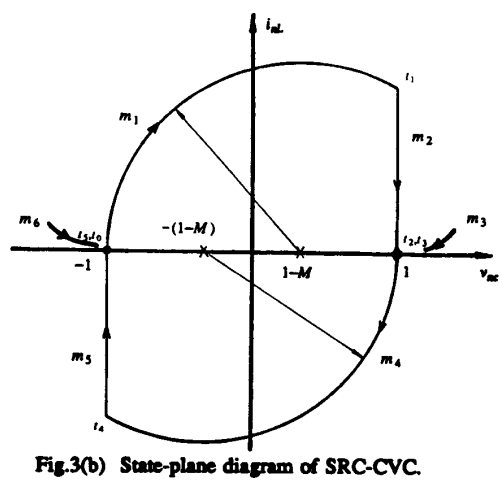

Fig.3(b) State-plane diagram of SRC-CVC.

The source V_g delivers energy to the tank and the load during modes m_1 and m_4 only. In modes m_2 and m_5 the load receives its energy from the inductor L. In modes m_3 and m_6 the load receives the energy from the output filter capacitor C_0. No energy is returned to the source in any circuit mode. Compared to the conventional SRC, the present converter has two following advantages,

(1) The peak source current is lower.

(2) There is no power loss in a practical voltage source due to the returned energy.

2.2 Design Parameters

The average power delivered by the source, normalized by V_g^2/Z_0 is given by,

$$P_{no} = \frac{2}{T_s}\int_{t_0}^{t_1} i_{nL}(t)\,dt = \frac{2}{\pi} f_{ns} \tag{9}$$

where the f_{ns} is the normalized switching frequency with respect to the resonant frequency.

By equating the power supplied by the source to the power delivered to the load , it can be shown that the relation among the converter gain M, the quality factor of the converter Q_l, and the normalized switching frequency f_{ns} is given by,

$$M = \sqrt{\frac{2}{\pi} Q_l f_{ns}} \tag{10}$$

where,

$$Q_l = \frac{R_L}{Z_0}$$

From the state-plane diagram we can see that the converter gain M should be less than 1.0 in order to maintain the discontinuous mode operation. The upper limit on the switching frequency to maintain the discontinuous operation can be shown to be,

$$f_{nsm} = \frac{\pi}{\alpha - tan(\alpha)} \tag{11}$$

where

$$\alpha = Cos^{-1}\left(-\frac{M}{2-M}\right)$$

The average load current is given by,

$$I_o = \frac{2}{\pi}\frac{f_s}{f_0}\frac{V_g}{M} \tag{12}$$

The peak values of the current and voltage in the switches and the rectifiers are given as below,

$$I_{swmax} = I_{dmax} = (2-M)\,V_g/Z_0 \quad ; \quad V_{swmax} = 2V_g \tag{13}$$

The peak switching current is smaller for higher converter gain. The maximum peak current in the transistor in the event of a short circuit at the output is limited to $2V_g/Z_0$.

2.3 Performance Characteristics

The characteristic curves of Fig. 4 depict the relation among the converter gain M, the normalized switching frequency f_{ns} and the load Q_l. At low values of M, the switching frequency f_{ns} changes by small amount to compensate for a large change in the load resistance. However, from equation (13) one finds that the low M results in large peak currents through the switching device. A design tradeoff is, therefore, necessary in the selection of the converter gain. The converter with low M requires the output transformer with small turns-ratio and high bandwidth. Thus the converter with SRC-CVC topology can be operated at high switching frequencies.

The converter gain M can also be regulated by controlling the resonant frequency f_0, pre-regulating the input voltage before applying as a source voltage to SRC-CVC or by clamping the diodes D_1 and D_2 to another voltage in place of V_g. The simplest control scheme, however, is the switching frequency control. The variation of the clamping level permits constant frequency operation.

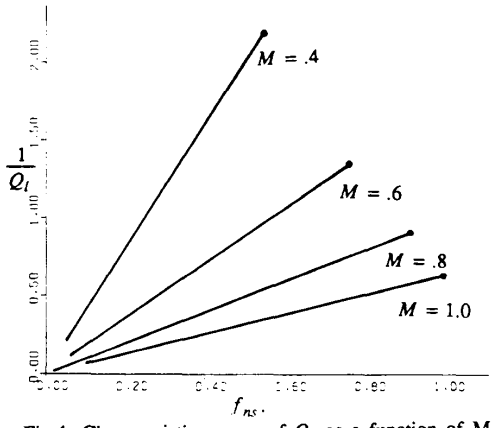

Fig.4 Characteristic curves of Q_l as a function of M and f_{ns}.

In this topology the switches are turned on and off at zero current. The rectifier diodes are switched off at zero current so there is no restriction of using the diodes with low reverse recovery time, such as the Schottkey diodes.

The SRC-CVC, like any other 2nd order SRC topology does not take into account the capacitances across the transformer windings and the switches S_1 and S_2. Thus inspite of low turns ratio transformer and the zero current turn-off in transistors and diodes, the operation at high frequencies is limited.

III. DESIGN OF A TWO-OUTPUT POWER SUPPLY

A two-output 100 watts power supply was designed using the SRC-CVC . The specifications of the supply are as follows,

input voltage	: 150 volts DC +/- 10%
output voltages	: 5 volts, max. load 10 amps.; 12 volts, max. load 4 amps.
output load regulation	: 5-volt and 12-volt outputs within .1% for a change in load from 50% to 100% of the maximum load current.
output ripple	: less than 5 mV p-p in 5-volt and 100 mV in 12-volt
switching frequency	: maximum frequency not less than 500 kHz and minimum frequency not less than 100 kHz.
power output	: 100 watts at efficiency greater than 80%.

The power stage and the control circuit for the supply are shown in Fig.5(a) and Fig.5(b), respectively. The design uses the MOSFETs IRF630 as switches and shottkey diodes, FE 8G as rectifiers. The maximum permissible peak current of 4 amp in the MOSFET and the +10% change in the input voltage limits our choice of M to be .8 . The output transformer has a turn ratio of 11 with 11 turns in the primary winding, 1 turn in the 5-volt and 3 turns in the 12-volt winding. The transformer is wound on ferrite E-core type 41515-EC . The leakage inductance of the transformer is 6μh and therefore no extra resonant inductor is required. The resonant capacitor is 13 nF. The resonant frequency is 570 kHz and maximum switching frequency is designed to be 524 khz. The characteristic impedence Z_0 of 21.5 ohms is obtained from eqn.(11). The supply has an efficiency well over 80%. The switching frequency is above 500 kHz at full load and drops to only 280 kHz at 50% of full load. Fig.6 gives the plot of the output power versus the switching frequency. It can be shown that the experimental values are close to the theoretical values over a wide range of the switching frequencies. The deviations at higher switching frequency can be attributed to higher copper losses.

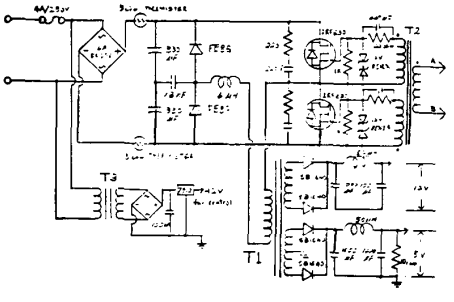

Fig.5(a) The power stage of the two-output power supply

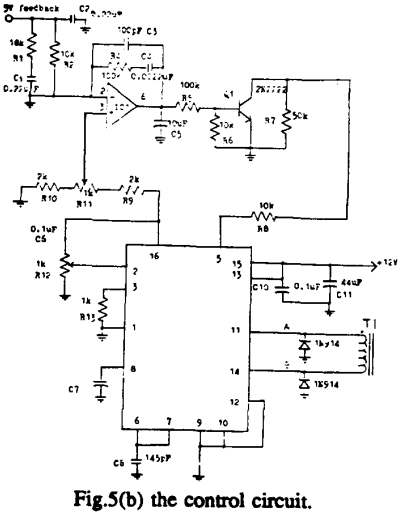

Fig.5(b) the control circuit.

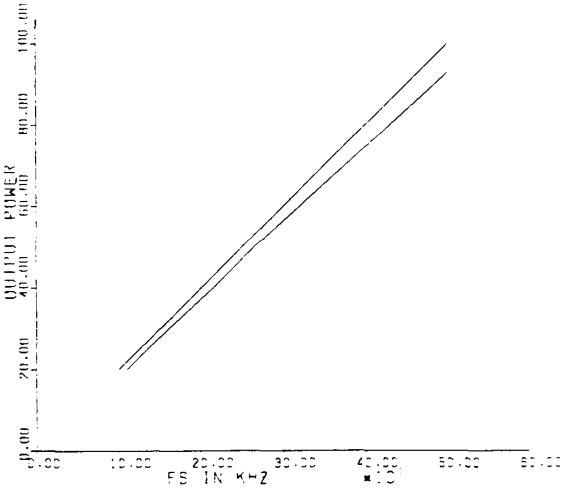

Fig.6 Output Power vs. Switching Frequency

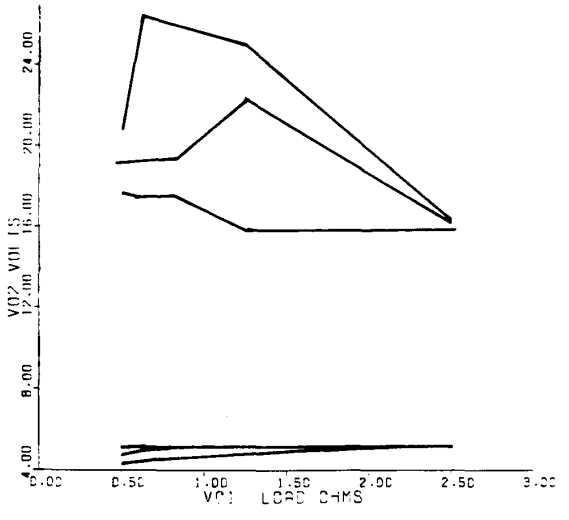

Fig.7 Cross regulation of 12-volt supply without nonlinear inductor

The 5-volt output is the main output and is regulated by varying the switching frequency. The unregulated 12-volt output changes even at its constant load when the load on the 5-volt is changed. Fig.7 shows the plot of 12-volt output versus the load change in the 5-volt supply. The change in the output voltage of the 12-volt supply is higher when its load current is much smaller than the load current in 5-volt supply and the source voltage is higher than its nominal value. This is the classic problem of cross regulation in multi-output power supplies. In the following section, a new method is presented to correct this problem.

IV. IMPROVED CROSS REGULATION USING NONLINEAR INDUCTOR

The cross regulation in a two-output power supply can be defined as the percentage change of voltage on one of the outputs for a specified load change on the other , while the output voltage of the later is held constant. It is caused by the unequal loading and unequal leakage inductances in the secondary windings of the output transformer [2]. One can minimize the difference in the leakage inductances by careful transformer design. The cross regulation problem due to unequal loading can be solved by isolating the two outputs for a fraction of a switching period . It is done by introducing a switch in series with the secondary winding of the unregulated output supply. The duration of the OFF state of the switch is controlled by the error signal which is obtained from comparing the unregulated output to the desired voltage.

Our design uses a nonlinear inductor to synthesize an equivalent switch connected in series with the secondary winding of the unregulated supply (V_{o2}). The high-inductance and low-inductance states of this inductor simulate respectively the off and on states of the proposed switch. The amount of inductance is controlled by a current which is derived from the error voltage in the control loop of the output V_{o2}. Fig.8(a) shows the simplified circuit of the two-output SRC-CVC incorporating the above switch. Since the unregulated 12-volt supply uses a center-tapped secondary winding it needs two nonlinear inductors L_1 and L_2, with the magnetic characteristics as given in Fig.8(b).

The steady state response of the modified converter can be analized by considering its operation over a switching period. At the beginning of a half switching period the diodes D_5 and D_6 are reverse biased .The inductor L_1 is biased at point B on the magnetic characteristic curve as shown in Fig. 8(b). It is in the

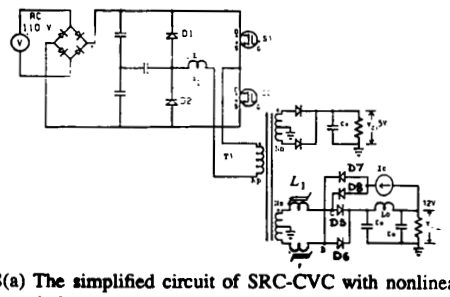

Fig.8(a) The simplified circuit of SRC-CVC with nonlinear inductor

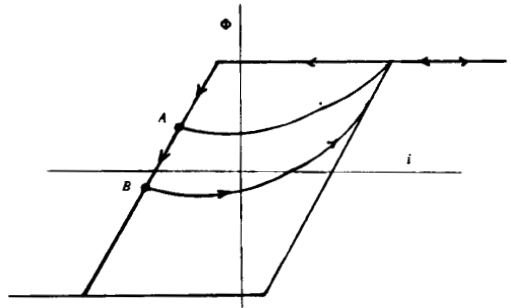

Fig.8(b) Magnetic characteristics of the nonlinear inductor

high-inductance state and virtually isolates the output V_{o2} from the primary of the transformer as well as the other output V_{o1} . The volt-seconds of the inductor L_1 brings it into saturation along the indicated path on magnetic characteristic curve as shown in Fig.8(b). After a time interval τ the inductance of L_1 drops abruplty and the source begins to supply power to the load of the 12-volt supply. During this half switching period, the inductor L_2 is in the high-inductance state. The diode D_7 conducts the reset current I_c to bias L_2 at the point B on the magnetic characteristic curve. In the next half switching cycle the inductor L_2 acts as a switch to isolate the 12-volt supply from the source and the other output. The inductor L_1 is reset to point B on the magnetic characteristic curve.

The isolation time τ of the outputs depends on the bias point B , which is controlled by the reset current I_c. Fig.9(a),(b) and (c) show the wave shapes of the currents in the primary and the two secondary windings of the transformer. The τ is the isolation time and the time interval $\tau'-\tau$ corresponds to the collapse of the nonlinear inductor from high-inductance state to the low-inductance state. The upper limit of τ is determined by the

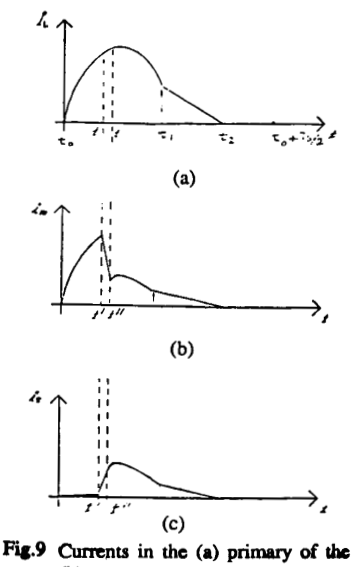

Fig.9 Currents in the (a) primary of the transformer (b) 5-volt output and (c) 12-volt output

maximum unsaturated flux density of the core material of the inductor which in turn limits the maximum peak current to the 12-volt load.

4.1 Experimental Results

The two inductors L_1 and L_2 are wound on ferrite core with narrow B-H characteristics, Toshiba amorphous core type MB 9X7X4.5 mm, with peak flux $\Phi_c AW$ of $73x10^{-8}$ Wb cm^2. Each inductor has 5 turns of wire AWG#19. The circuit to control the reset current in the nonlinear inductors is given in Fig.10. The IC, TL 431 provides a precision reference voltage with which the 12-volt output is compared. The error voltage controls the current through the transistor 2N4403 which resets the inoperative inductor to point B on the magnetic characteristic curve of Fig. 8(b).

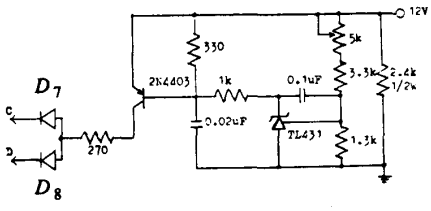

Fig.10 The Control Circuit for Reset current

Fig.11 shows the plots of the output voltages of 5-volt and 12-volt supplies with respect to the changing load resistance on the 5-volt supply. The cross regulation of the 12-volt supply has improved significantly. This method also regulates the 12-volt supply for the variation in its own load current and changes in the source voltage due to input line variations. Since the maximum value of τ is limited by the core material of the nonlinear inductor the extent of the load and cross regulations are limited and can be improved only at the expense of each other.

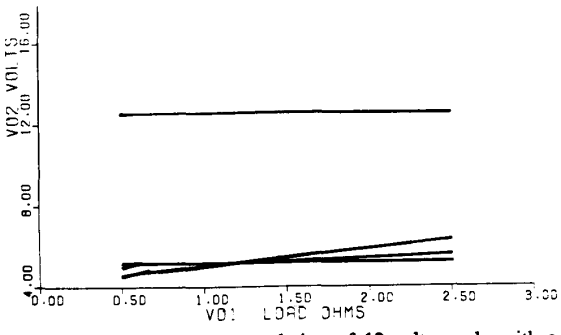

Fig.11 Improved cross regulation of 12-volt supply with a nonlinear inductor.

V. CONCLUSION

The SRC-CVC avoids any energy to be returned to the source so the component stresses are limited. The switching devices operate at zero-current off/on and the rectifiers commutate at zero current. The analysis and the performance curves show that this converter can be operated at low converter gain. The low converter gain makes it possible to use the output transformer with low turns-ratios and high bandwidth and thus enable the converter operation at high switching frequencies.

Using the above topology, a two-output power supply, capable of supplying 100 watts at greater than 80% efficiency, has been succesfully designed. The problem of cross regulation has been solved using a nonlinear inductor to isolate the two outputs over a part of the switching period. The result is a significant improvement in the cross regulation. The problem of cross regulation is being relatively sidetracked by the researchers while designing the post regulators for the multi-output power supplies. The cross regulation in fact puts an upper limit on the range of post load regulation.

Reference

[1] C.Q.Lee and K. Siri: analysis and design of series resonant converter by state-plane diagram. IEEE Trans. on AES-22, 757-763, Nov. 1986.

[2] Wilson T.G. Jr.: Cross regulation in an energy-storage DC-to-DC converter with two regulated outputs. Proc. of the IEEE PECS, Palo Alto, Cal., June 1977.

[3] Kwang H. Liu: Effects of Leakage Inductances on the Cross Regulation in a Discontinous-Mode Flyback Converter. Proc. of HFPC, Florida, May 1989.